

APPIAH ELLIOT RICHARD

+852 5191 0075 appiahelliott1@gmail.com

Interests

Computer / Electronic Engineering, Mathematics & Data Science

Core Competencies

- Python, MATLAB, JavaScript, Vue.js, HTML, CSS, React, Next.js, SQL, Supabase, PostHog
- Data Analysis, Research, UI/UX Design, Figma
- Da Vinci Resolve, Microsoft Suite, Leadership,
- Technical Writing, Mentorship, Problem-Solving, Team Collaboration

Education

The Hong Kong University of Science and Technology

Aug 2024 – Sep 2028

Electronic Engineering

Hong Kong

Mawuli Senior High School

Sep 2019 – Sep 2022

Valedictorian, WASSCE

Ho, Ghana

Mater Ecclesiae School

Jun 2017 – Sep 2019

Principal Honor Roll, BECE

Ho, Ghana

Work Experience and Research

Software Developer — Silvertrees Ventures Limited

Winter 2025

Roameo AI

Hong Kong

- Led a full UI/UX overhaul across layouts, components, and core user flows.
- Designed analytics pipeline with Supabase RPCs, PostHog funnels, and in-memory caching; reduced 20–30s rollups to <5s via SQL optimization + indexes.
- Integrated FastAPI service for ranked attractions and matching logic; built automated HTML email digests via Vercel cron + Resend

Full Stack Developer — HKUST TLE Program

Summer 2025

Sparrow

Hong Kong

- Designed and beautified core pages with Figma, Vue.js, and Vuetify improving visual consistency and mobile accessibility.
- Developed an interactive map search feature with Leaflet.js and Vue2-Leaflet, enabling students to explore 20-30 pre-filled housing listings with filterable categories for enhanced navigation.
- Prototyped comparison page using Vue.js, Axios, CSS Grid.
- Integrated Vue Router and Axios to manage seamless page navigation and data retrieval, ensuring a cohesive and dynamic user experience across the platform.

Video Producer/Content Creator/Writer

June 2025

HKUST Sustainability Office

Hong Kong

- Produced engaging videos/tutorials for sustainability campaigns.
- Used AI tools to create data-driven technical content.

Research Assistant / Program Coordinator

Ongoing

Pioneer Academics

USA (Remote)

- Serve as a pivotal liaison between world-class U.S. university professors and diverse international students, providing 6 hours of weekly mentorship, technical guidance, and coaching to drive innovative research over 12 weeks.
- Liaised with professors and mentored students weekly.

Curriculum Developer & Instructor STEMEX Academy, Hong Kong *Ongoing*

- Developed a comprehensive RoboMaster curriculum tailored for high school students in Hong Kong.
- Delivered engaging lessons on RoboMaster, Scratch, and Python to students, fostering technical skills and creativity.
- Prepared students for RoboMaster competitions, guiding them in applying their skills to real-world challenges.

Freelance Coding Instructor <i>Preface School of Technology</i> – Spearhead Python, Minecraft, Roblox, Scratch, App Inventor and web development courses, leveraging world class learning materials to deliver personalized, portfolio-ready projects like dynamic web apps and automation tools. – Mentored diverse learners and led workshops.	Ongoing <i>Hong Kong</i>
R&D Assistant <i>Voyager Education</i> – Developed robot/software modules and STEAM programs.	Feb 2024 <i>Hong Kong</i>
Web Developer/Programmer <i>Pet Pet Club Ltd.</i> – Revamped website and evaluated deep learning models.	Dec 2024 <i>Hong Kong</i>
Student Ambassador <i>HKUST</i> – Led campus tours, supported events and outreach.	Ongoing <i>Hong Kong</i>
Intern <i>Daystar Power Solutions</i> – Installed 4.3MW plant, saving 30% in energy/emissions.	Oct 2023 <i>Kumasi, Ghana</i>
Researcher <i>KNUST with Dr. Adu Boakye</i> – Built ambulance queuing model now used by 5 hospitals.	Jul 2023 <i>Kumasi, Ghana</i>
Awards and Honors • HKSAR Belt and Road Scholarship, HKUST Global Learner’s Award • Innovation Excellence Award, STEM Hero, NSMQ Valedictorian • Ghana Science Olympiad Gold Medalist, Tech-Up Africa Distinction • Dr. Adu Boakye Math/Physics/Chemistry Award	
Leadership Experience	
Founder <i>Elliot Explains</i> – Created animated science videos (170k+ views), supported 5 NSMQ teams.	2020 <i>YouTube</i>
Team Captain <i>National Science & Maths Quiz</i> – Led team to Top 9 nationally; secured learning center renovations.	2022 <i>Ghana</i>
President <i>National Sharks Quiz</i> – Grew club 5x, raised funds, planned weekly drills.	2022 <i>Ghana</i>
Co-Founder & VP <i>IT Hub</i> – Revamped website, built E-voting system.	2021 <i>Ghana</i>
Operations Manager <i>Kubi Foundation</i> – Managed clinic/borehole projects, empowered girls with tie-dye skills.	2022 <i>Ghana</i>
Founder <i>Kindred Hearts</i> – Led beach cleanup, recycling, and disaster donations.	2023 <i>Ghana</i>
Tutor <i>Aikins Educational Consult</i> – Tutored SAT/ACT Math and Science.	Aug 2023 <i>Accra, Ghana</i>

Academic Competitions

- NSMQ Regional Champion & Semifinalist, National Sharks Quiz Runner-up
- UCMAS Regional Champion, Ghana Science Olympiad Gold Medalist
- UG Numeric Nexus Winner, National Code Fest Winner

Personal Attributes

- High analytical proficiency, adept in data-driven projects.
- Calm under pressure with strong communication and leadership.
- Strong interpersonal and presentation skills.